

Recurrent Neural Network Based Classification

Assignment 5.4 - Deliverable Report

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1. Introduction

The goal of this experiment is to classify human activities (walking, sitting, standing, etc.) using **raw accelerometer and gyroscope signals** from smartphones. Three recurrent neural network architectures were evaluated: **RNN, LSTM, and GRU**. No feature engineering was applied; the raw signals were directly used.

The dataset is the **UCI HAR Dataset**, consisting of 7352 training samples with 128 timesteps and 6 sensor channels (3-axis accelerometer + 3-axis gyroscope).

2. Dataset Overview

- **Input shape:** (7352, 128, 6)
 - **Number of classes:** 6 (Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, Laying)
 - **Train/Validation Split:** 80% train, 20% validation
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3. Model Architecture

Three sequence models were implemented using PyTorch:

1. **RNN:** Vanilla recurrent network
2. **LSTM:** Long Short-Term Memory network
3. **GRU:** Gated Recurrent Unit network

Hyperparameters:

- Input size: 6
- Hidden size: 128
- Number of layers: 2
- Batch size: 64
- Epochs: 30
- Optimizer: Adam, LR=0.001

All models take raw sequences (128 timesteps × 6 features) as input and output class probabilities.

4. Training Results

4.1 Loss Curves

- **RNN:** Training and validation loss fluctuated; slow convergence.
 - **LSTM:** Training loss decreased initially but validation loss became unstable → overfitting.
 - **GRU:** Both training and validation loss decreased steadily and almost overlapped → stable learning and good generalization.
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4.2 Accuracy and F1-Score

MODEL	ACCURACY	F1-SCORE
RNN	36.85%	0.2435
LSTM	21.75%	0.0990
GRU	82.94%	0.8274

Observation: GRU outperforms RNN and LSTM significantly on raw data due to its gating mechanism that efficiently captures temporal dependencies.

4.3 Confusion Matrix (GRU Example)

- GRU accurately classifies activities like **Walking** and **Laying**, which have distinctive signal patterns.
 - Misclassifications occur between similar activities such as **Sitting** vs **Standing**, likely due to limited raw signal differences.
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5. Comparison with Fully Connected Neural Network (Deliverable 5.3)

- FCNN treats sequences as flat vectors → ignores temporal dependencies.
 - Previous FCNN experiments achieved ~**60–70% accuracy**, lower than GRU (~83%).
 - **Conclusion:** Recurrent models, especially GRU, are better suited for sequential sensor data.
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6. Observed Limitations

1. **Sensitivity to Raw Input:** RNN and LSTM struggle on unprocessed data. Preprocessing may improve performance.
 2. **Vanishing/Exploding Gradients:** RNN suffers from gradient issues over long sequences.
 3. **Training Time:** Recurrent models take longer than FCNNs to converge.
 4. **Overfitting Risk:** LSTM showed unstable validation loss → overfitting.
 5. **Interpretability:** Temporal patterns learned by RNN/LSTM/GRU are hard to interpret.
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7. Conclusions

- **GRU** is the best-performing model for HAR with raw sensor data.
- Vanilla **RNN** and **LSTM** have poor performance due to gradient and generalization issues.
- **Recurrent models** effectively capture temporal dependencies, outperforming FCNN on sequential data.
- For future work, **feature engineering, normalization, or hybrid CNN-RNN models** could improve performance further.